Double-Angle Formulas and Trigonometric Equations

ANSWER KEY

Double-angle formulas

1 Given that $\cos\theta = -\frac{3}{5}$ and θ lies in Quadrant II, find:

a $\sin 2\theta \;= -\frac{24}{25}$ **b** $\cos 2\theta \;= -\frac{7}{25}$ **c** $\tan 2\theta \;= \frac{24}{7}$

2 Use a double-angle formula to write $\cos^2 x$ in terms of $\cos 2x$, and state the corresponding formula for $\sin^2 x$.

$\cos^2 x = \frac{1+\cos 2x}{2}$ and $\sin^2 x = \frac{1-\cos 2x}{2}$.

Solving trigonometric equations

3 Solve $2\sin x - 1 = 0$ on the interval $[0, 2\pi)$.

$\sin x = \frac{1}{2}$, so $x = \frac{\pi}{6}$ or $x = \frac{5\pi}{6}$.

4 Solve $2\cos^2 x - \cos x - 1 = 0$ on the interval $[0, 2\pi)$.

Factor: $(2\cos x + 1)(\cos x - 1) = 0$. From $\cos x = -\frac{1}{2}$: $x = \frac{2\pi}{3}, \frac{4\pi}{3}$; from $\cos x = 1$: $x = 0$. Solution set: $\left\{0, \frac{2\pi}{3}, \frac{4\pi}{3}\right\}$.

5 Solve $\sin 2x = \sin x$ on the interval $[0, 2\pi)$.

$2\sin x\cos x - \sin x = 0$, so $\sin x(2\cos x - 1) = 0$. From $\sin x = 0$: $x = 0, \pi$; from $\cos x = \frac{1}{2}$: $x = \frac{\pi}{3}, \frac{5\pi}{3}$. Solution set: $\left\{0, \frac{\pi}{3}, \pi, \frac{5\pi}{3}\right\}$.

6 Solve $\tan^2 x = 3$ on the interval $[0, 2\pi)$.

$\tan x = \pm\sqrt{3}$, so $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$.